

PAPER_454 — MUGE Compression Cycle 2: 19-System Multi-Registry Expanded Gravitational Calculator

Whitepaper Series: Star-Magic UQFF Phase 2

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Classification: FIRST 19-system UQFF/MUGE compression registry; FIRST multi-class environmental compression from magnetar through cosmological scales

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CP4 Class: MultiSystemCompressionCycle2Calculator (#8, PAPER_454)

Abstract

Compression Cycle 2 expands the 7-system canonical registry (PAPER_452) to a 19-system multi-scale catalogue incorporating objects from neutron star surfaces ($r \sim 10^4$ m) to cosmological filaments ($r \sim 10^{26}$ m). The 12 newly added systems span HII regions, galaxy mergers, galaxy clusters, and the Hubble Ultra-Deep Field, each contributing system-specific F_{env} terms. The single compressed equation $g_{UQFF} = g_{Newton} \times (1 + H_z t) + F_{env}(t)$ applies uniformly across 22 orders of magnitude in spatial scale — demonstrating the universality of the MUGE compression framework for all observed astrophysical environments.

2. 19-System Registry — PAPER_454

2.1 Full System Catalog

The 19-system registry extends the 7-system base (PAPER_452) with 12 additional systems:

#	System	Type	M (kg)	r (m)	F_{env} dominant
1	MagnetarSGR1745	Neutron star	5.58×10^{30}	1×10^4	B-field + decay
2	SagittariusA	SMBH	8.17×10^{36}	6×10^9	Accretion disk
3	TapestryStarbirth	SFR region	9.96×10^{33}	1×10^{16}	SFR + outflow
4	Westerlund2	Star cluster	1.99×10^{34}	6×10^{16}	Stellar wind
5	PillarsCreation	HII pillars	3.98×10^{32}	6×10^{16}	Radiation
6	RingsRelativity	GL system	1×10^{39}	1×10^{20}	Lensing
7	UniverseGuide	Cosmological	1×10^{53}	4.4×10^{26}	F_{cosmo}
8	NGC2525	Barred spiral	$\sim 2 \times 10^{41}$	$\sim 5 \times 10^{20}$	Tidal arm
9	NGC3603	Massive cluster	$\sim 4 \times 10^{34}$	$\sim 3 \times 10^{17}$	OB wind
10	BubbleNebula	Wind bubble	$\sim 1 \times 10^{32}$	$\sim 2 \times 10^{17}$	Stellar bubble
11	AntennaeGalaxies	Merger	$\sim 4 \times 10^{40}$	$\sim 2 \times 10^{21}$	Tidal merger

#	System	Type	M (kg)	r (m)	F_env dominant
12	Horsehead Nebula	Bulard 33	$\sim 5 \times 10^{31}$	$\sim 5 \times 10^{15}$	B-field + PDR
13	NGC1275	Perseus cluster	$\sim 1 \times 10^{43}$	$\sim 3 \times 10^{22}$	Cluster ICM
14	NGC1792	Spiral galaxy	$\sim 1 \times 10^{41}$	$\sim 4 \times 10^{20}$	Disk wind
15	HubbleUDF	Deep field	$\sim 10^{53}$	$\sim 4 \times 10^{26}$	Statistical ensemble
16	StudentsGuide Nebula	Perseus	variable	variable	All terms
17	Lagoon Nebula	M8 HII region	$\sim 5 \times 10^{33}$	$\sim 1 \times 10^{17}$	Radiation + ionisation
18	Trifid Nebula	M20 tri-furcated	$\sim 1 \times 10^{33}$	$\sim 8 \times 10^{16}$	Radiation + dust
19	Omega Nebula	M17 Swan	$\sim 3 \times 10^{33}$	$\sim 1 \times 10^{17}$	O-star radiation

2.2 Compressed MUGE Equation (Universal Form)

$$g_{\text{UFF}}^{(j)}(t) = \underbrace{\frac{GM_j}{r_j^2}(1 + H_z t)(1 - B_j/B_{\text{crit}})}_{g_{\text{Newton, Hubble, B}}} + \underbrace{\sum_k U_{gk}^{(j)}}_{U_{g1} + U_{g2} + U'_{g3} + U_{g4}} + \underbrace{F_{\text{env}}^{(j)}(t)}_{\text{system-specific}}$$

2.3 New F_env Types (Systems 8-19)

Tidal merger (Antennae Galaxies):

$$F_{\text{env, tidal}} = \frac{GM_1 M_2}{(d_{12})^3} \cdot r$$

Where d_{12} = separation between merging cores, r = field evaluation point.

ICM (NGC 1275 Perseus cluster):

$$F_{\text{env, ICM}} = \frac{kT_{\text{ICM}}}{\mu m_H r_{\text{cool}}} = \frac{1.38 \times 10^{-23} \times 5 \times 10^7}{0.62 \times 1.67 \times 10^{-27} \times 3 \times 10^{22}} \approx 2.7 \times 10^{-12} \text{ m/s}^2$$

With $T_{\text{ICM}} \approx 5 \times 10^7$ K (Perseus cooling flow).

Stellar wind bubble (NGC 3603/Bubble Nebula):

$$F_{\text{env, bubble}} = \frac{\dot{M}_{\text{wind}} v_{\text{wind}}}{4\pi r_{\text{bubble}}^2}$$

OB stellar wind (Lagoon/Trifid/Omega):

$$F_{\text{env, OB}} = \frac{L_{\text{OB}}}{4\pi r^2 c} = P_{\text{rad, OB}}$$

3. Scale Universality of MUGE Compression

The 19 systems span 22 orders of magnitude in radius and 22 orders in mass:

Property	Min	Max	Range
Radius (m)	10^4 (magnetar)	4.4×10^{26} (universe)	22 dex
Mass (kg)	5.6×10^{30} (magnetar)	10^{53} (universe)	22+ dex
g_{UQFF} (m/s ²)	$\sim 10^{-34}$ (ultra-low)	$\sim 10^{12}$ (magnetar surface)	46 dex

The **single compressed equation** handles the full range with only F_{env} changing between systems. This is the UQFF universality principle: **the gravitational equation is scale-free**.

4. Total System Gravity Budget at t=1 Gyr

$$G_{\text{total}}^{(19)} = \sum_{j=1}^{19} g_{\text{UQFF}}^{(j)}$$

Dominated by Magnetar surface gravity:

$$G_{\text{total}}^{(19)} \approx 3.73 \times 10^6 \text{ (Magnetar)} + O(10^0 \text{ to } 2) \text{ (others)} \approx 3.73 \times 10^6 \text{ m/s}^2$$

For normalised comparison (setting $g_{\text{Magnetar}} = 1$ reference):

System	g / g_{Magnetar}
Magnetar	1.0
SgrA* (at $r=6 \times 10^9$ m)	3.0×10^{-7}
Galaxy clusters	$\sim 10^{-15}$
Universe	$\sim 10^{-21}$

5. Standard Model Comparison

Feature	SM	CC2-19 System
Multi-class gravity	Separate codes per system	Unified registry
Scale coverage	Different equations at different scales	Single g_{UQFF} for 22 dex
ICM coupling	Separate X-ray / hydro	F_{env} , ICM built-in

Feature	SM	CC2-19 System
Tidal mergers	N-body codes	F_env,tidal in registry

6. Testable Predictions

1. **Scale-free universality:** $g_{\text{UQFF}} \propto GM/r^2$ for all 19 systems to leading order, with $F_{\text{env}}/g_{\text{Newton}} < 10^2$ for all systems — testable by comparing UQFF output at each system's characteristic radius.
2. **ICM temperature coupling:** $F_{\text{env,ICM}} \propto T_{\text{ICM}}$ — doubling Perseus cluster temperature to 10^8 K should double the F_{env} contribution. Testable via Chandra X-ray spectra.
3. **Tidal merger timing:** $F_{\text{env,tidal}}$ for Antennae grows as d_{12} decreases — UQFF predicts $F_{\text{env}} \propto d_{12}^{-3}$ increasing by $8\times$ as separation halves. Observable in VLBI proper-motion measurements.

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$	$L_X L \geq 10^{37} \text{ erg/s}$	Chandra CXC	✓ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	✓ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite *PAPER_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

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